Fiscal Year:	FY 2019	**Task Last Updated:**	FY 11/19/2019
PI Name:	Riaz, Amir Ph.D.		
Project Title:	Gravity Scaling of Pool Boiling Heat Transfer: Numerical Simulations and Validation with MABE and NPBX		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ariaz@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-0707
Organization Name:	University of Maryland		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	127 Glenn L Martin Hall		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	08/16/2016	**End Date:**	08/18/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 8/18/2020 per NSSC information (Ed., 11/12/19) NOTE: Extended to 8/15/2019 per NSSC information (Ed., 10/3/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kim, Jungho Ph.D. (University of Maryland) Balaras, Elias Ph.D. (The George Washington University)		
Grant/Contract No.:	NNX16AQ77G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The key challenge in developing two-phase thermal systems for space environments is the development of a heat transfer database and reliable models for boiling in variable gravity environments from which the performance of two-phase heat exchangers can be predicted. The Microheater Array Boiling Experiment (MABE) and Nucleate Pool Boiling eXperiment (NPBX) were flown on the International Space Station (ISS) in 2011 within the Boiling Experiment Facility (BXF). MABE provided local heat transfer and visualization of pool boiling heat transfer behavior in a microgravity environment. Earth gravity data obtained prior to launch served as the reference data. The ability of a model developed from parabolic aircraft work and ISS data to predict how boiling was altered under a range of gravity levels was tested and found to predict the ISS data within 20%. The focus of NPBX was the investigation of how single and multiple bubbles (up to 5) grew and merged in microgravity. Heat fluxes for steady state nucleate boiling and critical heat flux were found to be much lower than those obtained under Earth gravity. It was found that heater size, chamber geometry, and bulk fluid conditions must be accounted for in nucleate boiling heat transfer correlations under microgravity conditions. In the proposed work, high fidelity numerical simulations of the boiling process will be used to glean further insights into the boiling process. The data from MABE and NPBX will be used to validate the simulations as they provide a wide range conditions (heater sizes, subcoolings, and superheats including CHF) at which to test the numerical results. NPBX will serve as validation for simulations with a few nucleation sites in microgravity. Simulations of boiling in Earth gravity will be made for the MABE heater and compared with the MABE Earth gravity results, after which additional simulations will be performed as gravity is slowly decreased. The results will be compared with the MABE microgravity results and the model. Co-Investigators for the proposed work are Prof. Elias Balaras (George Washington University) and Prof. Jungho Kim (U. of Maryland, PI for MABE).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reliable, physics based direct numerical simulation techniques that are also numerically efficient will be used to interrogate the fundamental physics of the boiling process under low-gravity and Earth-gravity conditions. Such high accuracy modeling of the boiling process will help develop effective heat transfer methodologies for the cooling of various engineering equipment including microprocessors.
Task Progress:	Over the last one year, we have made progress in reproducing MABE experiments and quantifying statistical behavior of bubble dynamics at different gravity levels. Quantitative and qualitative validations with analytical, experimental, and published numerical results have been carried out to demonstrate the accuracy and robustness of the solver. A quasi-random nucleation site model, based on low discrepancy Halton sequence is implemented to match nucleation site density available from experimental data by Kim et al. (2002). Pool boiling simulations results show that gravity has a strong effect on wall heat transfer and leads to regime change in terms of bubble shape, size, and quasi-steady behavior. The results match experimental data and trends. The simulations provide a good data set to analyze bubble dynamics for low gravity and transition regimes, which can now be confidently undertaken using the present solver. Reference: Jungho Kim, John F Benton, and Derek Wisniewski. Pool boiling heat transfer on small heaters:effect of gravity and subcooling.International Journal of Heat and Mass Transfer, 45(19):3919– 3932, 2002.
Bibliography Type:	Description: (Last Updated: 04/23/2026)
Abstracts for Journals and Proceedings	Dhruv A, Balaras E, Riaz A, Kim J. "Direct numerical simulation of gravity effects on pool boiling heat transfer." 71st Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Atlanta, Georgia, November 18-20, 2018. Bulletin of the American Physical Society. 2018 Nov;63(13):BAPS.2018.DFD.G09.7. http://meetings.aps.org/Meeting/DFD18/Session/G09.7 , Nov-2018
Articles in Peer-reviewed Journals	Dhruv A, Balaras E, Riaz A, Kim J. "A formulation for high-fidelity simulations of pool boiling in low gravity." International Journal of Multiphase Flow. 2019 Nov;120:103099. Available online 4 September 2019. https://doi.org/10.1016/j.ijmultiphaseflow.2019.103099 , Nov-2019
Papers from Meeting Proceedings	Dhruv A, Balaras E, Riaz A, Kim J. "High-fidelity simulations of flow boiling." NURETH-18: 18th International Topical Meeting on Nuclear Reactor Thermal Hydraulics, Portland, Oregon, August 18-23, 2019. NURETH-18: 18th International Topical Meeting on Nuclear Reactor Thermal Hydraulics, Portland, Oregon, August 18-23, 2019. , Aug-2019